Fiscal Year:	FY 2010	**Task Last Updated:**	FY 11/02/2009
PI Name:	Nelson, Gregory A. Ph.D.		
Project Title:	Epigenetic Control of Radiogenic Damage Processing in C. elegans		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation Biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	(1) **BMed**:Risk of Adverse Cognitive or Behavioral Conditions and Psychiatric Disorders		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	92350-1700	**Congressional District:**	31
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2007 Space Radiation NNJ07ZSA001N
Start Date:	01/01/2008	**End Date:**	12/31/2011
No. of Post Docs:	0	**No. of PhD Degrees:**	1
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA ARC
Contact Monitor:	Cucinott1a, Francis	**Contact Phone:**	281-483-0968
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Flight Program:			
Flight Assignment:	NOTE: Received NCE until 12/31/2011, per NSSC info (10/2010)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Braby, Leslie (Texas A&M University) Ford, John (Texas A&M University)		
Grant/Contract No.:	NNX08AC79G		
Performance Goal No.:			
Performance Goal Text:			
	Levels of expression of specific genes in individuals are determined by the particular combination of alleles that they carry and the regulatory environment of cells expressing those genes. Molecular pathways and mechanisms that detect and respond to damage from ionizing radiation are thus subject to modulation by external processes acting upon the genes expressed in the irradiated cells. These mechanisms constrain and determine the final levels of gene activity. These epigenetic mechanisms may be cell autonomous or operate at the tissue and organism level by processing information from cytokines, growth factors, hormones, and extracellular matrix. The range of gene expression patterns driven by the combinatorial actions of these epigenetic regulatory mechanisms ultimately determines the inter-individual variability of responses to radiation exposure. We propose to address these issues using a simple animal model, the nematode C. elegans, in which genotoxicity (chromosome aberrations) can be quantified in vivo in individual intestine		

Task Description:	cells whose tissue location and cell lineage are invariant. We have shown that individual cells in this 20-cell tissue respond to genotoxic damage in their neighbors and that their radiosensitivities appear to be modulated by signals originating in an adjacent tissue, the immature gonad. To identify the mechanisms underlying these cell-cell and tissue-tissue interactions we will employ a targeted screening protocol based on RNA interference methods to determine which of approximately 250 candidate genes alter the tissue architecture, cell-cell communication properties, DNA repair capacity, and important regulatory signal cascades which we believe are likely to modulate the expression and transmission of the genotoxic damage. This will be complemented by targeted irradiations of individual cells and cell groups with microbeam and collimated alpha particles to determine the range and topology of the damage transmission between cells.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Low doses of ionizing radiation in the form of natural radioactivity, cosmic rays and medical diagnostic procedures comprise the background exposure levels for the general population. Occupational exposures may add to this dose and the sum of the two will establish health risks from radiation exposure. Risks from radiation exposure have been extrapolated from data on atomic bomb survivors, individuals exposed during radiation accidents, and older unregulated medical exposures. Regulatory limits have employed the linear no-threshold model for extrapolating risks to low doses but recent experimental findings suggest significant deviations from linearity at low doses such that the uncertainties in risk estimates must be re-evaluated. One of the important types of biological responses that may explain the deviations are those termed non-targeted effects. These are responses in which cells not directly exposed to radiation may sustain injury due to perturbations in their environment or transfer of damaging chemical species created in the irradiated cells. This task is focussed directly on identifying the genes and pathways that regulate non-targeted damage formation. The high genetic homology between C. elegans and mammalian gene networks should enable predictions of homologous signaling pathways that are operational in mammals, including humans. This should, in turn, improve the understanding of the role of non-targeted effects in the low radiation dose environment and suggest mitigation strategies.
Task Progress:	Management The grant “Epigenetic Control of Radiogenic Damage Processing in C. elegans” was approved by NASA as submission 07-RadStep2-0036 and assigned grant # NNX08AC79G. The effective period is for 36 months: 01/01/2008 through 12/31/2010. The principal investigator, G. Nelson and principal laboratory assistant, T. Jones have traveled to Texas A&M University several times since the original site visit to set up the laboratory and have conducted a series of "dry runs" in preparation for exposure of larvae. Procedures for mounting animals them in the microbeam and configuring the optics have been established but new methods relying on epi-fluorescent illumination are still under development. Nelson and Jones have received safety orientations and are now qualified for unescorted access to the microbeam facility. Graduate student Daniel Kesack, Jr. in the Nuclear Engineering Department at TAMU conducted a number of C. elegans experimental procedures in support of the project but has now completed his degree. Ms. Smantha Edgington, an undergraduate in Radiological Health Engineering, is now providing support to Drs. Ford and Braby for nematode experiments. Preliminary results for the project have been presented at three international meetings: 17th Int. C. elegans Meeting. 6/24-28/09 UCLA; Heavy Ions in Therapy and Space Symposium 2009. 7/6-10/09. Cologne, Germany; Radiation Research Society. Savannah, GA 10/4 – 7/09. There are no contractual or financial issues to report. Technical At Texas A&M University. We previously reported that the Van de Graaf accelerator suffered a serious leak resulting in loss of sulfur hexafluoride insulating gas. The leak has since been found, repaired and the system recharged with sulfur hexafluoride. During this accelerator “down time” we conducted several dry runs, testing the experimental procedures needed to accomplish the proposal goals, but without irradiation. These were very useful in that team members at Texas A&M have learned basic nematode procedures and supporting equipment has been set up to maintain and manipulate the nematodes on site in close proximity to the beam port. Several deficiencies in the imaging procedures and equipment intended to use for worm cell targeting were identified. While sufficient for larger adult worms, the smaller larval stages could not be imaged with sufficient resolution to insure proper dose delivery to individual gut cells, as required. This was addressed in several ways. We rebuilt the optical stage of the Zeiss Axiomat microscope - cleaned and upgraded the optics, developed a new custom mount for the microscope objective and microbeam particle detection sensors, and are installing a new higher resolution video imaging system to go with the new optics. We can now adjust a longer working distance objective through a greater focal range which has significantly improved the imaging and the camera upgrade should improve the resolution further. The Loma Linda team has developed and tested a transgenic worm strain with GFP expression for gut cell visualization and evaluated a neutral red staining technique that also increases gut contrast. To use this GFP strain, we are installing a blue LED epi-illumination capacity to the workstation. We are currently testing these techniques and will soon have the ability to target the beam with the required accuracy. Tentative plans for December 2009 irradiation experiments are being made. At Loma Linda. Progress has been made in all biological areas of the project. A retrospective analysis of over 50 irradiation experiments using photons, charged particles from protons to niobium ions, neutrons and alpha particles has been completed using statistical methods to describe interactions between individual cells of the worm intestine. These have revealed interaction patterns which appear to be independent of radiation quality. The RNAi and mutant screens have revealed requirements for non-homologous end joining DNA repair and unusual interactions with telomere processing pathways but little dependence on homologous recombination, “9-1-1” or other DNA repair pathways. So far, RNAi and mutation - based inhibition of innexin gap junction genes, Notch, Ras, JNK, MAP kinase, and ATM signal transduction pathway components have not disrupted either the putative gonad to gut or intestinal cell-cell interactions. However, overall radiosensitivity levels do show effects. Results presented at the 2009 international C. elegans meeting have allowed us to refine our gene selection strategy especially with regard to micro RNA control pathways and gap junction expression patterns. To deal with the microbeam optics issues we initiated a transgenic animal strain construction to tailor the expression of green fluorescent protein to the intestine in order to refine cell targeting at the TAMU microbeam facility where a complementary optics upgrade is in progress. A strain expressing GFP on an elt-2 gut-specific promoter has been

	constructed This was complemented by a vital dye staining method to further enhance contrast for targeting at the microbeam facility.
Bibliography Type:	Description: (Last Updated: 03/26/2026)
Abstracts for Journals and Proceedings	Jones T, Perez C, Ortloff L, Nelson GA. "Intercellular Transmission of Genotoxic Damage in the Nematode Intestine." 17th International C. elegans Meeting, University of California, Los Angeles, June 24 – 28, 2009. Conference Proceedings. 17th International C. elegans Meeting, University of California, Los Angeles, June 24 – 28, 2009. , Jun-2009
Abstracts for Journals and Proceedings	Jones T, Perez C, Ortloff L, Nelson GA. "In vivo Bystander Effects in the Nematode Intestine. " Heavy Ions in Therapy and Space Symposium 2009 and 20th Annual NASA Space Radiation Health Investigators' Workshop, Cologne, Germany, July 6-10, 2009. Conference Proceedings. Heavy Ions in Therapy and Space Symposium 2009 and 20th Annual NASA Space Radiation Health Investigators' Workshop, Cologne, Germany, July 6-10, 2009. , Jul-2009
Abstracts for Journals and Proceedings	Jones T, Perez C, Ortloff L, Nelson GA. "In vivo Bystander Effects in the Nematode Intestine. " Radiation Research Society 55th Annual Meeting. Savannah, GA, Oct. 4 – 7, 2009. Conference Proceedings. Radiation Research Society 55th Annual Meeting. Savannah, GA, Oct. 4 – 7, 2009. , Oct-2009